

Muon Studies of Superconductors

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Keywords

vortex lattice, time-reversal symmetry breaking

Abstract

Understanding superconductivity in its myriad forms arising in numerous different crystal architectures is one of the major quests of modern condensed matter physics. One promising avenue to gain local information about novel superconductors is the use of local probes to measure properties inside the unit cell. The application of muon spin spectroscopy to the study of various superconducting materials is reviewed. These experiments can be carried out as a function of temperature, magnetic field, and pressure and even in thin-film samples. They provide information about proximal magnetic phases and the nature of the superconducting state, as well as giving intriguing evidence of time-reversal symmetry breaking. To properly interpret the experimental results, it is necessary to have reliable information about the site of the implanted muon, as well as its stability. This can now be provided using density functional theory techniques.

1. INTRODUCTION

Spin-polarized positively charged muons (μ^+) have emerged as a powerful tool for studying superconductors (SCs). Muons are produced in accelerators via pion decay (see **Figure 1a**), which results in fully spin-polarized beams of muons if muons are extracted from pions stopped in the surface of the pion target (the resulting particles are known as surface muons). Most of the studies involve the positive muon μ^+ , which is implanted in the sample of interest, occupying a particular crystallographic site. The implantation process involves slowing the muon down from a speed of $\approx c/4$ and dumping the 4 MeV of kinetic energy with which muons emerge from pion decay. The slowing-down process is extremely rapid, taking place in ~ 1 ns, and crucially involves only Coulombic degrees of freedom so that the muon spin polarization is preserved. The technique is a by-product of work in fundamental particle physics and followed from the realization that muon decay ($\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$) involves parity violation (1), enabling the measurement of the asymmetry of positron (e^+) emission to facilitate a measurement of the time dependence of the implanted muon's spin polarization (see **Figure 1b**). This forms the fundamental basis of the muon spin rotation (μ SR) technique (2–5).

Following implantation, the subsequent time dependence of the muon spin polarization $\mathbf{P}(t)$ is determined by its magnetic interaction with the sample, with any local static magnetic field resulting in spin precession, because

$$\frac{d\mathbf{P}}{dt} = \gamma_\mu \mathbf{P}(t) \times \mathbf{B}(t). \quad 1.$$

A muon has a magnetic moment of 3.18 times the proton magnetic moment, or about $4.8 \times 10^{-3} \mu_B$, and hence its spin precesses in an applied magnetic field B at a frequency of $\omega = \gamma_\mu B$, where $\gamma_\mu/(2\pi) = 135.5 \text{ MHz T}^{-1}$. If the local magnetic fields fluctuate, this can result in spin relaxation, which can also be studied.

The mean lifetime of the muon is $\tau_\mu = 2.2 \mu\text{s}$, but because the lifetime of any individual muon follows an exponential probability distribution, it is possible to measure a significant number of muon decays that occur around ten times this value following implantation, if sufficient muons are initially implanted. Consequently, the muon spin polarization can be followed typically up to $\sim 20 \mu\text{s}$ after implantation, although the data become noisier at late times because

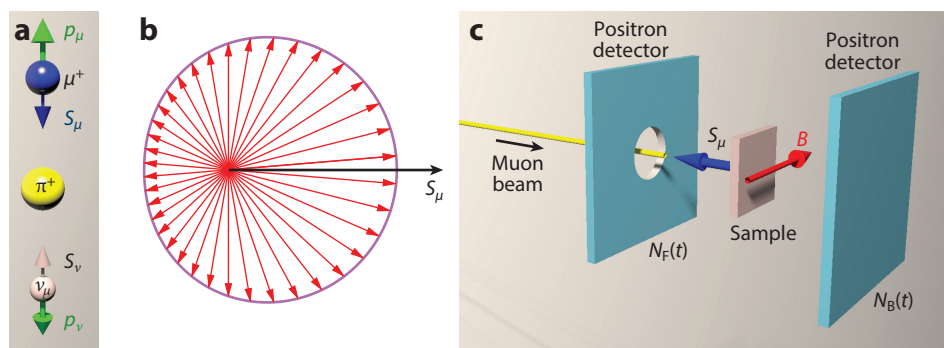


Figure 1

(a) A pion decays into a muon and a neutrino. The neutrino has its spin antiparallel to its momentum, so the muon has to do the same. In this way, 100% spin-polarized beams can be produced. (b) The angular distribution of emitted positrons with respect to the initial muon spin direction. (c) Schematic of a μ SR experiment, with the magnetic field applied in the transverse direction (a geometry often used for studying superconductors). Abbreviation: μ SR, muon spin rotation.

fewer and fewer muons survive in the sample as time increases. Detecting this precession is key to the μ SR technique, because it gives us a method of measuring the magnetic field B inside the sample. μ SR experiments have an obvious application to magnetically ordered materials, because spin precession occurs at a frequency proportional to the magnetic order parameter (6). For this reason, μ SR is used in many studies of superconducting phase diagrams in which there is a proximal magnetic phase. Experiments on samples in the superconducting state are often performed with a magnetic field applied transverse to the initial muon polarization, as shown in **Figure 1c**. The field measured at the muon site is inhomogeneous due to the way in which an external field penetrates the sample, and the μ SR technique gives a means for extracting that distribution (7).

I begin in Section 2 by describing how low-energy muons, that is, muons whose kinetic energy have been reduced well below the 4 MeV energy of surface muons, can be used to study surfaces of type I SCs. This is not the usual experiment that is performed, but it is probably the simplest one to describe. More usually, surface muons are used to focus on type II SCs, with the distribution of precession signals used to infer the magnetic penetration depth and provide information about the structure of the vortex lattice. This aspect is reviewed first, in Section 3, after which I turn to the study of magnetism in Section 4. I then discuss some technical issues concerning the determination of the site of the implanted muon in Section 5. Then, time-reversal symmetry breaking is considered in Section 6, in which a fundamental property of certain SCs may become manifest in microscopic magnetic fields, whose origin has been somewhat mysterious.

2. SUPERCONDUCTIVITY AT SURFACES

Although accelerators can readily produce 4-MeV muon beams, in order to study surfaces it is desirable to find a means of slowing these muons down. It turns out that layers of several hundred nanometers of rare gas solids can be used to moderate muon beams, preserving their spin polarization (8). The conversion efficiencies of the best moderators are in the 10^{-5} – 10^{-4} range, with solid neon having the highest efficiency, although solid argon tends to be used in practice because, although its conversion efficiency is about half that of solid neon, the muon beam causes sputtering of the thin neon layers in which the atoms are weakly bonded, and this seems not to be so much of a problem with argon. The active volume of the moderator is simply set by the depth from which the moderated muons can escape into the vacuum. The technique has been developed at the Paul Scherrer Institute in Switzerland to produce a moderated muon rate of over 10^4 s^{-1} (with an incoming surface muon rate of $1.9 \times 10^8 \text{ s}^{-1}$). Crucially, the facility is equipped with the ability to accelerate the moderated muons up to 20 keV by biasing the moderator substrate. This produces a low-energy muon beam with variable energy E that goes from a few hundred electron volts up to 20 keV, allowing depth-dependent studies to be performed because the muons can be implanted at a chosen average depth $\bar{z}(E)$ (though with a spread of depths $\Delta z(E)$, and both $\bar{z}$ and Δz are functions not only of E but also of the sample material). Helpfully, $\bar{z}(E_{\text{max}})$ slightly exceeds 100 nm, so it is the same order of magnitude as the penetration depth in high- T_c SCs; hence, superconducting-induced field variations can be probed (9). Distinct measurements for each value of $\bar{z}(E)$ reveal spin precession at an average Larmor frequency of $\omega(z) = \gamma_\mu B(z)$. The data shown in **Figure 2** are consistent with the exponential decrease with depth of the local field, thereby demonstrating the Meissner effect. Two sets of data are shown in **Figure 2b** because the single crystal sample of orthorhombic $\text{YBa}_2\text{Cu}_3\text{O}_{6.92}$ is an anisotropic SC (10), and the average local field as a function of energy is related to an integral of the field profile in the SC multiplied by the stopping profile of surface muons moderated with that energy. Similar methods have been used to study screening at the surface of superconducting Nb (11).

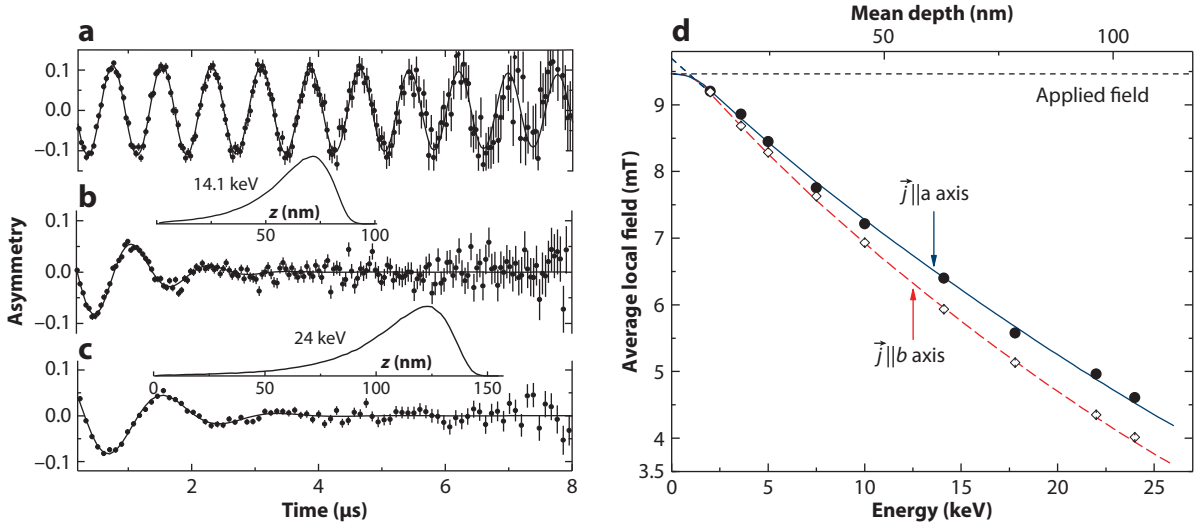


Figure 2

Low-energy muon data for $\text{YBa}_2\text{Cu}_3\text{O}_{6.92}$. (a) Data for $T > T_c$ show no relaxation because all muons see the applied field. (b,c) Below T_c the precession frequency is averaged over the depth distributions and decreases with increasing average depth $\bar{z}$, i.e., with increasing implantation energy. (d) The depth-dependent average field can therefore be extracted for many such runs, and from these data the penetration depth can be extracted. Figure adapted with permission from Reference 10; copyright 2010 American Physical Society.

3. THE SUPERCONDUCTING VORTEX LATTICE

When an applied magnetic field $\langle B \rangle$ is applied to a type II SC, it is not uniformly distributed throughout the SC but predominates in vortex lines, each one of which carries a flux quantum ϕ_0 . Thus, although the magnetic flux density takes an average value $\langle B \rangle$, it takes a larger value at the vortex cores and a smaller value between the vortex cores. Muons stopping close to a vortex core will therefore precess much more rapidly than those stopping halfway between the cores. Thus, the superconducting phase can be studied using μSR for the simple reason that a vortex lattice in a type II SC results in an inhomogeneous magnetic field distribution inside the material.

For a hexagonal vortex lattice, the distance between vortices, d , is such that the number of vortices per unit area $2/(\sqrt{3}d^2)$ equals the number of flux quanta per unit area B/Φ_0 . Thus, $d \propto B^{-1/2}$. (For a square vortex lattice of side $d_\square$, the number of vortices per unit area is $1/d_\square^2$, and the relationship $d_\square \propto B^{-1/2}$ holds.) In general, the vortex lattice is incommensurate with the crystal lattice and, except at very high magnetic field, the vortex cores are separated by a much larger distance than the unit-cell dimensions. In a transverse-field μSR experiment, muons are implanted with their initial spins polarized along the x direction, with the magnetic field B applied along the z direction. The implanted muons sit at certain crystallographic sites and, thus, randomly sample the field distribution of the vortex lattice. The field distribution $p(B)$ inside the SC is defined by $p(B) = \langle \delta(B - B(\mathbf{r})) \rangle_r$ and is the probability that a randomly chosen point in the sample has field B (see **Figure 3**). In general, the transverse-field muon spin relaxation function $P_x(t)$ is related to the field distribution $p(B)$ using

$$P_x(t) = \text{Re} \langle \exp(i\gamma_\mu B(\mathbf{r})t) \rangle_r = \text{Re} \left(\int_{-\infty}^{\infty} p(B) e^{i\gamma_\mu B t} dB \right), \quad 2.$$

and thus $P_x(t)$ can follow changes in $p(B)$ as both the temperature and applied field B_{applied} are varied.

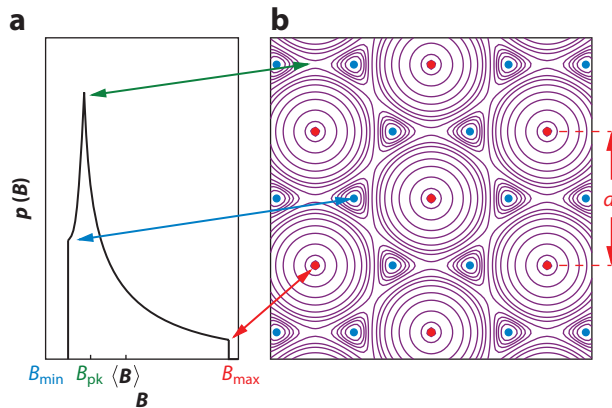


Figure 3

(a) Field distribution $p(B)$ in the superconducting vortex lattice. (b) A contour diagram of the field inside a hexagonal vortex lattice.

3.1. Vortex Lattice Field Distribution and Extraction of the Penetration Depth

The flux lattice in a type II SC is usually hexagonal, so that vortices occupy positions $p\mathbf{a}_1 + q\mathbf{a}_2$, where p and q are integers, and the basis vectors are $\mathbf{a}_1 = d(\sqrt{3}/2, 1/2)$ and $\mathbf{a}_2 = d(0, 1)$, where d is the spacing between adjacent vortices (i.e., the side of the triangle in the vortex lattice, so that the cell area is $\sqrt{3}d^2/2$). Fourier transforming this two-dimensional real-space lattice results in a two-dimensional lattice in reciprocal space, with reciprocal lattice vectors given by $\mathbf{G}_{mn}$ with integers m and n , where

$$\mathbf{G}_{mn} = \frac{2\pi}{d} \begin{pmatrix} m \\ \frac{1}{\sqrt{3}}(2n + m) \end{pmatrix}, \quad 3.$$

so that $G_{mn}^2 = [16\pi^2/(3d^2)](m^2 + n^2 + nm)$ as shown in **Figure 4a**. We can also define the magnetic field $\mathbf{B}(\mathbf{r})$ at position $\mathbf{r}$ within the SC as

$$\mathbf{B}(\mathbf{r}) = \sum_{\mathbf{G}} \mathbf{b}(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}}, \quad 4.$$

where $\mathbf{b}(\mathbf{G})$ is the Fourier transform of $\mathbf{B}(\mathbf{r})$, given by

$$\mathbf{b}(\mathbf{G}) = \frac{B}{\phi_0} \int_{\text{cell area}} \mathbf{B}(\mathbf{r}) e^{-i\mathbf{G}\cdot\mathbf{r}} d^2r, \quad 5.$$

which is an integral over the cell area $\sqrt{3}d^2/2 = \phi_0/\langle B \rangle$. The average field $\langle B \rangle$ is then given by

$$\langle B \rangle = \frac{B}{\phi_0} \int_{\text{cell area}} \mathbf{B}(\mathbf{r}) d^2r = \mathbf{b}(\mathbf{0}), \quad 6.$$

so that it is determined simply by the Fourier transform at $\mathbf{G} = \mathbf{0}$, and hence the variance in the field ΔB^2 is given by

$$\Delta B^2 = \langle (B - \langle B \rangle)^2 \rangle = \langle B^2 \rangle - \langle B \rangle^2 = \sum_{\mathbf{G} \neq \mathbf{0}} b^2(\mathbf{G}). \quad 7.$$

The London equation for SCs can be written as

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda_L^2}, \quad 8.$$

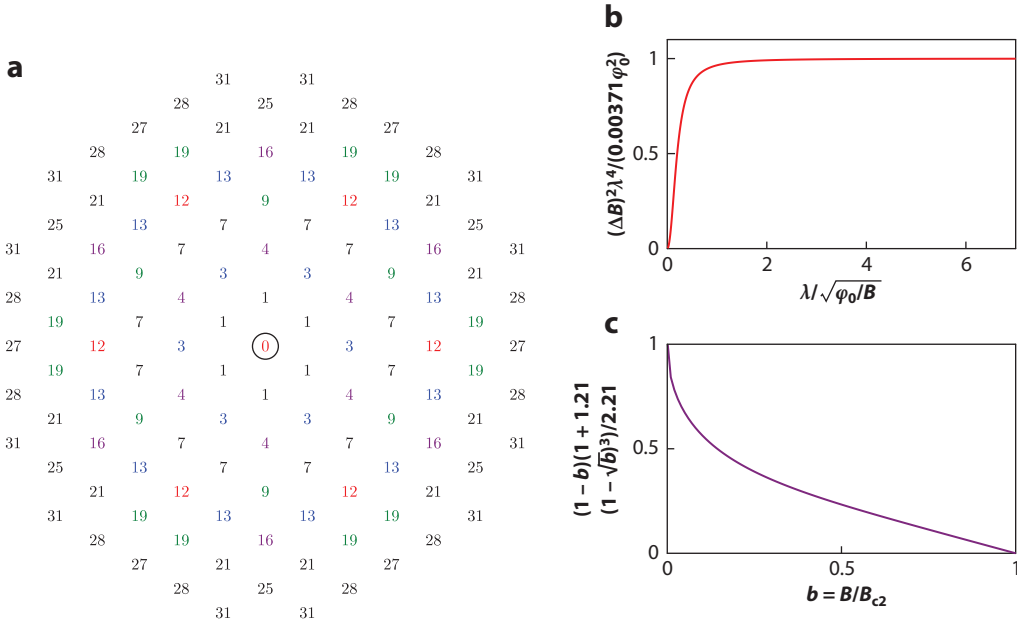


Figure 4

(a) The reciprocal lattice of the hexagonal flux lattice, with the points labeled by the value of $G^2_{mm}/[16\pi^2/3d^2]$, which is useful for evaluating Equation 13. (b) Equation 13, which yields $\Delta B^2 = 0.00371 \frac{\phi_0^2}{\lambda^4}$, relies on the condition $G\lambda \gg 1$, and (c) using instead Equation 16 shows the field dependence.

where λ_L is the London penetration depth. For the case of the vortex lattice, this becomes

$$\lambda_L^2 \nabla^2 \mathbf{B} - \mathbf{B} = \phi_0 \hat{\mathbf{z}} \sum_i \delta^{(2)}(\mathbf{r} - \mathbf{r}_i), \quad 9.$$

where the sum runs over all the vortices at positions $\mathbf{r}_i$. Fourier transforming this yields

$$\mathbf{b}(\mathbf{G}) = \frac{\langle B \rangle \hat{\mathbf{z}}}{1 + G^2 \lambda^2}, \quad 10.$$

and, hence,

$$\Delta B^2 = \sum_{\mathbf{G} \neq 0} \frac{\langle B \rangle^2}{(1 + G^2 \lambda^2)^2}. \quad 11.$$

In the case that $G\lambda \gg 1$, this becomes

$$\Delta B^2 \approx \frac{\langle B \rangle^2}{\lambda^4} \sum_{\mathbf{G} \neq 0} \frac{1}{G^4} = \frac{\langle B \rangle^2}{\lambda^4} \sum_{\substack{m,n \\ (m,n) \neq (0,0)}} \frac{1}{G_{mn}^4}. \quad 12.$$

This is a sum over points in the reciprocal lattice (12), as shown in **Figure 4a**, and by evaluating the sum (with the help of that diagram) this leads to

$$\begin{aligned} \Delta B^2 &= \frac{3\phi_0^2}{64\pi^4 \lambda^4} \times 6 \times \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{2}{7^2} + \frac{1}{9^2} + \frac{1}{12^2} + \frac{2}{13^2} + \dots \right) \\ &= 0.00371 \frac{\phi_0^2}{\lambda^4}. \end{aligned} \quad 13.$$

This result is used extensively in μ SR. Note that it relies on the condition $G_{10}\lambda \gg 1$ being satisfied; this equates to $8\pi^2\langle B\rangle\lambda^2/(\sqrt{3}\phi_0) \gg 1$ (see **Figure 4b**). Taking the lower critical field $B_{c1} = (\phi_0/4\pi\lambda^2)(\frac{1}{2} + \ln\kappa)$, where $\kappa = \lambda/\xi$, then for the case $\kappa \gg 1$, the condition becomes

$$\frac{\langle B \rangle}{B_{c1}} \gg \frac{\sqrt{3}}{2\pi \ln \kappa}, \quad 14.$$

meaning that the applied field $\langle B \rangle$ has to be significantly larger than B_{c1} . The above treatment has also ignored the finite size of the vortex core, but this can be included (13).

Experimentally, ΔB^2 is extracted by fitting the transverse-field data $P_x(t)$ with an appropriate relaxation function and subtracting a nuclear-dipole contribution σ_n , which is assumed to be temperature independent and is determined from the $T \gg T_c$ data. The superconducting contribution is then

$$\langle \Delta B^2 \rangle_{sc} = \langle \Delta B^2 \rangle - \sigma_n^2, \quad 15.$$

which assumes that the nuclear and superconducting contributions add in quadrature. Note that $\sqrt{\langle \Delta B^2 \rangle_{sc}}$ represents the standard deviation of the field distribution and is, thus, a measure of the width of the field distribution plotted in **Figure 3a**. Although there are other techniques for extracting the penetration depth, an advantage of μ SR is that data are obtained from the bulk of the SC, in contrast to techniques involving microwaves, which are only sensitive to effects within a skin-depth of the surface.

3.2. Magnetic Field Dependence

Using Ginzburg–Landau theory, the field dependence of the contribution to the superconducting width can be written in terms of the reduced field $b = B/B_{c2}$ using

$$\langle \Delta B^2 \rangle_{sc}^{1/2} = \frac{A(1-b)[1 + 1.21(1 - \sqrt{b})^3]}{\lambda^2} (\mu\text{m}^{-2}), \quad 16.$$

a function that is plotted in **Figure 4c**. The constant A in this equation takes the value 4.83 and 5.07 for hexagonal and square flux line lattices, respectively (14), and this can be tested with experiment (15, 16).

If the gap function has line nodes, the Volovik effect can lead to magnetic-field-induced quasi-particle excitations around gap nodes (17), and this can lead to the density of excited states increasing proportionally to $\sqrt{B}$, resulting in an extracted penetration depth following $\lambda(b)/\lambda(0) = (1 - K\sqrt{b})^{1/2}$, where K is a constant expressing the strength of the Volovik effect; this can also be tested in experiments on samples with nodes in the gap function (18).

3.3. Relation to Band Structure

The zero-temperature value of the in-plane penetration depth is related to the underlying electronic band structure, so that for a two-dimensional SC it can be written as a sum over contributions from the i th band via

$$\frac{1}{\lambda_{ab}^2(0)} \equiv \sum_i \frac{1}{\lambda_{ab,i}^2(0)} = \frac{e^2}{2\pi\epsilon_0 c^2 b L_c} \sum_i \oint v_{F,i}(\mathbf{k}) d\mathbf{k}, \quad 17.$$

where L_c is the c -axis lattice constant, $v_{F,i}(\mathbf{k})$ is the Fermi velocity of the i th electronic band at momentum $\mathbf{k}$, and the integrals are taken around the Fermi surface (19). This has allowed measurements of the penetration depth to be obtained from the Fermi surface properties found using angle-resolved photoemission spectroscopy (ARPES) and then to be directly compared with those

obtained from μ SR (20). The temperature dependence of the superfluid density $\rho_s(T)$ can be related to the temperature dependence of the superconducting gap function $\Delta(T)$ via

$$\rho_s(T) = \frac{\lambda^{-2}(T)}{\lambda^{-2}(0)} = 1 + \frac{1}{\pi} \int_0^{2\pi} d\phi \int_{\Delta(\phi)}^{\infty} \left(\frac{\partial f(E, T)}{\partial E} \right) \frac{E dE}{\sqrt{E^2 - \Delta^2(T, \phi)}}, \quad 18.$$

where $f(E, T) = [1 + \exp(E/k_B T)]^{-1}$ is the Fermi function, and this allows models of different superconducting gap functions to be tested experimentally (see, for example, 21–24). One possible parameterization is to write the gap function $\Delta(T, \phi) = \Delta(T)g(\phi)$ with a temperature dependence given by $\Delta(T) = \Delta(0)\tanh\{1.82[1.018(T_c/T - 1)]^{0.51}\}$ (25). The angular dependence $g(\phi)$ would then be given by $g(\phi) = 1$ or $\cos(2\phi)$ for an s-wave or d-wave gap, respectively, and in this way different gap symmetries can be tested. In favorable cases (particularly when the penetration depth is quite short, so that ΔB^2 is large), these can be distinguished using μ SR, though sometimes there are several different models of the gap symmetry that are all reasonably consistent with the data.

3.4. Vortex Lattice Physics

In highly anisotropic systems the vortex lattice is no longer a system of rigid rods but should be considered as a system of flexible interacting lines. A useful picture is that of a weakly coupled stack of quasi-two-dimensional (q2D) “pancake” vortices, each one confined to a superconducting plane (26, 27). The phase diagram is thus substantially altered to take account of field and temperature dependent changes in the vortex lattice itself. At low T and low B , the stacks resemble conventional vortex lines. Above a characteristic temperature T_b , but still below that at which superconductivity is destroyed, the vortex lattice is broken up by thermal fluctuations (27) (this is called vortex lattice melting). At low T , but this time at increasing B , the energetic cost of interlayer deformations of the lattice (local tilting of the lines) is progressively outweighed by the cost of intralayer deformations within the superconducting plane (shearing). Above a crossover field B_{cr} , the vortex lattice enters a more two-dimensional regime. Thus in anisotropic systems, one may expect temperature- and field-dependent transitions in which the vortex lattice is destroyed, and all of these can be studied using μ SR (28, 29).

3.5. Vortex Lattice Motion

μ SR can also be used to study the field distribution inside moving vortex lattices, driven by a current (due to the fact that individual vortex lines experience a Lorentz force $\mathbf{F}_L = \mathbf{J} \times \mathbf{B}$ per unit volume, where $\mathbf{J}$ is the current density), as long as the Lorentz force exceeds the pinning force due to the effect of any quenched disorder. In this case, the muon spin autocorrelation function $P_x(t)$ follows

$$P_x(t) = \text{Re} \left\langle \exp \left(i \gamma_\mu \int_0^t B(\mathbf{r} - \mathbf{v}t') dt' \right) \right\rangle_{\mathbf{r}}, \quad 19.$$

where $\mathbf{v}$ is the velocity of the vortex lattice. Experiments of this type in Pb-In give results that are in agreement with small-angle neutron scattering (30) and can provide microscopic information on flux flow (31).

3.6. The Uemura Plot

The London formula for the zero temperature limit of the penetration depth $\lambda(0)$ yields in the clean limit (the mean free path much bigger than the coherence length) the following relation: $\lambda(0) \propto \sqrt{m^*/n_s(0)}$, where m^* is the effective mass and $n_s(0)$ is the density of superconducting

electrons. This can be combined with measurements of the Sommerfeld constant to yield a value for the Fermi temperature T_F . Thus, from muon measurements it is possible to plot a diagram showing the relationship between the Fermi and critical temperatures for a range of SCs (32, 33). This picture has come to be called a Uemura plot and shows a clear correlation between T_c and T_F for the heavy fermion, organic, fullerene, and Chevrel phase SCs and is equivalent to a scaling $T_c \propto \lambda^{-2}$. The conventional, elemental SCs lie away from this correlation and have values of $T_c/T_F \sim 10^{-3}$ (for the exotic SCs, this value is one or two orders of magnitude larger). This correlation has been interpreted as evidence that the exotic SCs may be close to Bose–Einstein condensation of preformed local pairs, which is expected to occur at a temperature of $\sim T_F$ (34). This is somewhat related to Homes scaling (35) and may be violated in organic SCs, although this has been debated (36, 37). Uemura interprets his plot in terms of a model of a resonance between spin and charge energy scales, and applies this concept to explain the unusual behavior in the overdoped regime of high- T_c cuprates in which the transition temperature is suppressed even though the Fermi temperature continues to increase (38).

4. MAGNETISM AND SUPERCONDUCTIVITY

The primary use of μ SR in studying magnetism is particularly useful for many new SCs because they often exhibit interesting magnetic properties in addition to their superconducting properties. Sometimes this is because both magnetism and superconductivity coexist in the same sample, a feature of some multifunctional materials that then allows μ SR to be extremely useful in demonstrating that both properties coexist at a microscopic level (39). More commonly, the magnetic phase is proximal to superconductivity with the undoped material exhibiting an antiferromagnetic Mott insulator state and superconductivity obtained by doping the Mott insulator (40).

4.1. Cuprates

This situation was of course realized by the high- T_c cuprates, and it was notable that μ SR first established the magnetism in undoped samples (41). Part of the reason for this is that μ SR is an extremely efficient technique for measuring a series of samples, and phase diagrams can be quickly explored and magnetic phases easily identified. The connection between magnetism and superconductivity is believed to be important because the pairing mechanism is not dominated by the electron–phonon interaction [as for conventional Bardeen–Cooper–Schrieffer (BCS) SCs] but is instead mediated by electronic spin fluctuations; the μ SR data give clear evidence for a non-s-wave gap (42, 43). As might be expected, there is an extensive literature of μ SR studies in this area, which has been reviewed elsewhere (7, 44, 45). In common with other techniques, more recent studies have benefitted from progress in sample quality. μ SR is particularly effective in studying the magnetism in lightly doped compounds (46). μ SR has also been used to rule out the presence of static magnetic moments in the pseudogap regime (47, 48). Very recently, it has provided evidence for spin-density wave order at ambient pressure in the double-layer nickelate $\text{La}_3\text{Ni}_2\text{O}_{7-\delta}$ (49), a compound that exhibits high-temperature superconductivity under pressure and appears to demonstrate that cuprate-like superconductivity can occur in nickel-containing materials (50).

4.2. Heavy Fermion Superconductors

In heavy fermion materials, the large effective mass is believed to result from the strong correlations between the conduction electrons and the localized f moments. Superconductivity is observed in a number of heavy fermion systems, often coexisting with static magnetism, probably resulting from the fact that the attractive effective interaction between electrons is provided

by spin fluctuations. μ SR studies have therefore been extremely important in characterizing the physical properties of heavy fermion systems (51).

4.3. Iron-Based Superconductors

The discovery of superconductivity in iron-based materials catalyzed a number of muon studies that quickly established the basic phase diagrams (52–57) in which superconductivity was, as for the cuprates, found in the undoped phases with superconductivity appearing on charge doping (see **Figure 5a,b**). An example of this approach is shown in **Figure 6a**, which shows μ SR data for $\text{NaFe}_{1-x}\text{Co}_x\text{As}$. Two precession signals are clearly visible for the undoped material, but these become suppressed on doping (58, 60). By comparing the structural (obtained from synchrotron

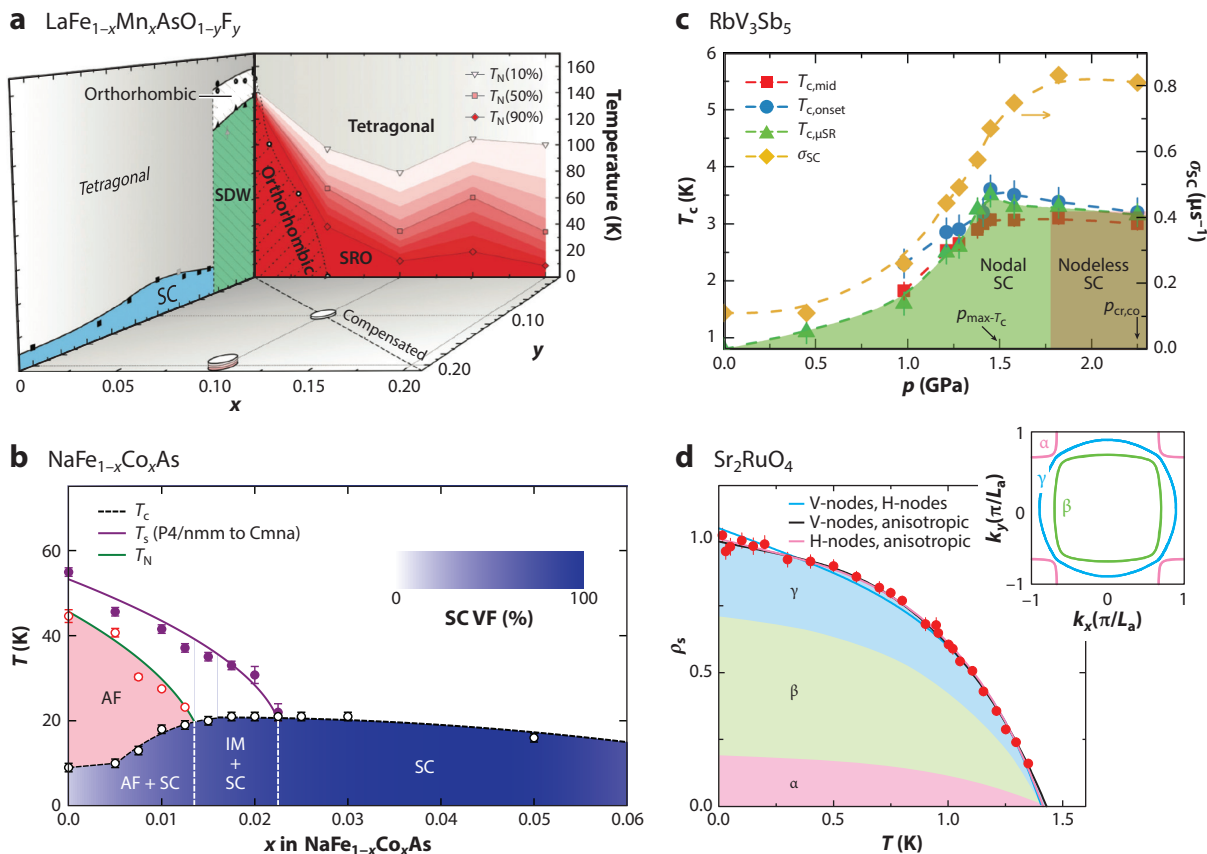


Figure 5

Phase diagrams for a few families deduced from μ SR studies. (a) $\text{LaFe}_{1-x}\text{Mn}_x\text{AsO}_{1-y}\text{F}_y$, obtained by fluorine substitution on the oxygen site and Mn substitution on the Fe site. (b) $\text{NaFe}_{1-x}\text{Co}_x\text{As}$, obtained by cobalt substitution on the Fe site, deduced from the data shown in **Figure 6**. This compound shows a structural transition between a tetragonal phase with space group $P4/nmm$ to an orthorhombic phase with space group $Cmma$. (c) RbV_3Sb_5 , obtained as a function of pressure. (d) The temperature dependence of the superfluid stiffness in Sr_2RuO_4 , with the different colored regions representing the contributions from the α , β , and γ regions on the Fermi surface (shown in the *inset*). Panel a adapted from Reference 57; copyright 2013 American Physical Society. Panel b adapted with permission from Reference 58. Panel c adapted from Reference 24 (CC BY 4.0). Panel d adapted with permission from Reference 59. Abbreviations: μ SR, muon spin rotation; AF, antiferromagnetism; IM, inhomogeneous magnetism; SC, superconductor; SDW, spin-density wave; SRO, short-range order; VF, volume fraction.

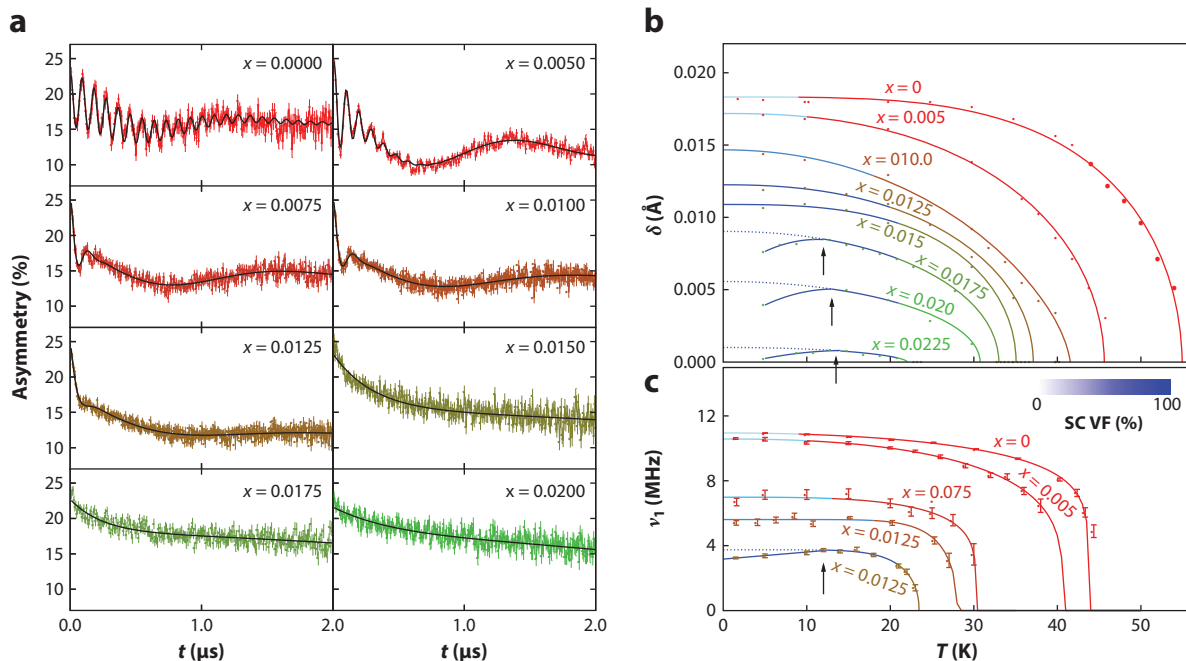


Figure 6

(a) Comparison of the zero-field μ SR asymmetry spectra for $\text{NaFe}_{1-x}\text{Co}_x\text{As}$, measured around 1.5 K. (b) The evolution of the structural order parameter (δ is the difference between the a and b lattice parameters in the low-temperature Cmma phase) and (c) the magnetic order parameter (ν_1 is the highest of the two observed muon precession frequencies in the zero-field data in panel a), as a function of temperature T and x . The blue line sections show where the sample superconducts, and the shade of blue indicates the volume fraction obtained from magnetometry. Figure adapted with permission from Reference 58. Abbreviations: μ SR, muon spin rotation; SC, superconductor; VF, volume fraction.

X-ray data; see **Figure 6b**) and magnetic (obtained from μ SR; see **Figure 6c**) order parameters, it is apparent that they follow different temperature dependences, with magnetism setting in below the structural transition (and leading to the phase diagram shown in **Figure 5b**). Both the magnetism and the structural distortion are weakened as the robust superconducting state is established at low temperatures (**Figure 6b,c**), which can be interpreted in terms of a model (61) involving coupled nematic, magnetic, and superconducting order parameters (similar results are found for Ni doping of NaFeAs) (21).

Although chemical substitution can be used to tune these materials, introduction of spacer layers (62) and intercalation can also be used (63) to push iron-chalcogenide layers into new configurations. μ SR can be performed under hydrostatic pressure (64), and this has allowed the pressure dependence in various iron-based SCs, such as FeSe , to be studied (65–68). There have also been various calculations of the muon site in iron-based SCs (21, 65, 69), which can help in quantitative interpretation of experimental data.

4.4. Kagome Lattice Superconductors

New SCs based on the kagome lattice of corner-sharing triangles have become of strong recent interest. The tight-binding model on the kagome lattice contains Dirac points and flat bands, and the resulting electronic instability of this band structure facilitates both the formation of charge-density waves and unconventional superconductivity (70). μ SR has been used to study materials

of the form AV_3Sb_5 , where A is an alkali atom (usually K, Rb, or Cs) (24, 71, 72) (see **Figure 5c**) and also $LaRu_3Si_2$ (73). This work has been reviewed in Reference 74.

4.5. Strontium Ruthenate

Sr_2RuO_4 is a multiband SC with a layered perovskite structure that has attracted enormous interest, owing to the difficulty in finding a model of the order parameter that is consistent with all experiments, which remains elusive. The μ SR evidence for time-reversal symmetry breaking is discussed in Section 6. Its phase diagram has been obtained as a function of uniaxial stress (75), using a specially designed piezoelectric-driven uniaxial pressure cell that is optimized for muon spin relaxation measurements (76). The contributions to the superfluid stiffness from different sections of its multiband Fermi surface are shown in **Figure 5d** (59).

5. MUON SITES

In transverse-field studies of the superconducting vortex lattice, we may not be primarily concerned with precisely where the muon stops inside the sample. It is enough to know that it is at some crystallographic site, and then any observed field variation comes from the vortex lattice is, in general, incommensurate with the crystal lattice, ensuring that the muon samples this field distribution uniformly. However, it is often important to consider the final stopping site of the muon, particularly if the focus is the nature of local magnetism. If there is relatively little contact hyperfine coupling, the stopped muon will resemble something like a bare particle acting as an interstitial defect, albeit dressed by interactions with the electronic system. This might take the form of the muon acquiring a screening cloud of electronic charge in a metal, or being associated with a strain field in an insulator as it deforms atoms in its vicinity. In metals, the muon is likely to occupy an interstitial site (in face-centered cubic copper, it goes for the octahedral interstitial site) (77), but in ionic compounds the positive muon favors sites close to negatively charged anions (for example, in oxides, including cuprate SCs, the muon sits around 1 Å from an O^{2-}) (78), with the resulting state behaving like a hydroxyl OH group, which has a bond length close to 1 Å.

5.1. Locating the Muon Site

Muon sites can now be determined with a high degree of accuracy using density functional theory (DFT) techniques. The so-called DFT+ μ method (79–83) begins by randomly assigning an initial muon site in a unit cell of the target material and relaxing the structure, by allowing both the muon and the atoms in the crystal to adjust their positions via repeated iteration until the forces on them are reduced below some threshold. The total energy of the final relaxed configuration is then evaluated, and the process is repeated for several other randomly chosen initial muon sites, with the lowest total energy of the relaxed structure identifying the most likely muon site, as well as providing an estimate for the structural distortion the muon introduces. Experiments at continuous sources of muons are performed with one muon in the sample at any one time (the muon decays before the next muon is likely to arrive in the sample); experiments at pulsed sources of muons may have 10^2 – 10^3 muons in a pulse, all arriving together, but even here the number of muons in the macroscopic sample (containing probably well over 10^{21} atoms) is much smaller than the number of atoms, so we never need to worry about the interaction between muons. The DFT codes most commonly used assume periodic boundary conditions, so the structure is usually specified via a supercell, containing several unit cells together with a single muon. The charge state of the muon is determined by the overall charge of the defect (+1 for diamagnetic and neutral, zero for paramagnetic states), which is fixed initially; note that for the diamagnetic case, a uniform, smeared-out Hartree-like compensating charge is added to the supercell to ensure

overall neutrality, and the final charged state of the impurity is to be determined by the final relaxed configuration (79). The muon therefore sits in a potential well, but it is important to note that its zero-point energy will be sizable, owing to the lightness of the muon ($m_\mu \approx m_p/9$), raising the possibility that the muon can hop or tunnel out of the potential well.

5.2. Diffusion of the Muon

Finding the muon site is important, but equally important is knowing under what conditions this site is stable or whether it hops between sites (although if the hop rate were $\lesssim 10^3 \text{ s}^{-1}$, the muon would be effectively static on the timescale of any experiment). The muon is initially self-trapped by its own local distortion of the lattice, but a tunnelling transition to another site is possible via phonon-induced fluctuations of the lattice. Thus, thermally-assisted diffusion of the muon through the lattice becomes increasingly likely at high temperature. This depends, of course, on the nature of the material. For example, in face-centered cubic copper, the muon is found to be static at an octahedral site (due to dipole broadening from nearby nuclear moments) with hopping detected above 100 K (77). In contrast, for face-centered cubic aluminum the muon is found to be mobile down to very low temperature, though the introduction of impurities has been used to introduce traps in which the muon can become localized (84, 85).

At low temperatures, the muon is therefore expected to be static, but experimentally it is found that in some materials the muon hop rate falls to a minimum on cooling and then begins to rise again. In the very low temperature regime, the muon can be in a quantum delocalized state for which phonons now appear to be hindering hopping rather than helping it, mainly because they are responsible for inelastic scattering, which destroys the coherence of this delocalized state (86, 87). The smaller mass of the muon leads to larger tunnelling matrix elements to neighboring sites and, thus, enhances the quantum mechanical nature of the motion over and above what would be expected for a hydrogen impurity.

In metals, the conduction electrons screen the muon and cannot react fast enough to the diffusing particle and cannot follow it adiabatically. This produces a net drag, which reduces the particle hop rate (88). If the metal undergoes a transition to superconductivity, the presence of the gap in the electronic spectrum effectively decouples the electron bath from the muon. Superconductivity can be removed by applying a sufficiently large magnetic field to the sample, reconnecting the muon diffusion process with the electron bath, increasing the drag and reducing the muon diffusion rate, as demonstrated in experiments on aluminum (89) and also tantalum (90).

In oxides, the muon is expected to be much more strongly bound and less likely to diffuse. It tends to form an OMu state, analogous to a hydroxide ion (91). Because this state does not possess an unpaired electron, it does not give rise to hyperfine interactions and so should precess in external or internal magnetic fields with the frequency of the bare μ^+ . This state is often imagined as being entirely immobile, and this must frequently be the case; if it were not, it would not be possible to see the multiple frequencies that are often observed in oxides from several coexisting sites [examples include the occurrence of six precession frequencies due to the charge disproportionation in AgNiO_2 (92) or the two frequencies in LiVGe_2O_6 (93)]. Nevertheless, there are examples of cases in which the muon state is delocalized over several such states (94, 95). Furthermore, a study on the cubic perovskite KTaO_3 (96) has highlighted the possibility of muon diffusion in this material (and possibly many other oxides) due to a relatively low activation energy ($\sim 0.1 \text{ eV}$), which arises from the large zero-point energy of the muon.

6. TIME-REVERSAL SYMMETRY BREAKING

Many of the recently discovered SCs exhibit unconventional superconductivity that does not follow the predictions of the BCS theory (97). A crucial issue in resolving the mechanism for

TRSB: time-reversal symmetry breaking

unconventional superconductivity is the presence or absence of time-reversal symmetry breaking (TRSB) (98), a property that can provide a tight constraint on the symmetry of the superconducting gap. The conventional s-wave singlet BCS pairing preserves time-reversal symmetry, but triplet pairing may not in some cases. For example, a p-wave gap symmetry was ascribed (99, 100) to the SC Sr_2RuO_4 on the basis of its supposed triplet order parameter, deduced from nuclear magnetic resonance (NMR) (101) and the presence of TRSB deduced from μSR (102), as well as from polar Kerr effect (103) measurements. However, the triplet nature of Sr_2RuO_4 has recently been discounted on the basis of a new NMR investigation (104), thereby reopening the question about the nature of the gap symmetry in this compound, with several alternative singlet gap structures proposed (75, 105–107), each of which would be consistent with TRSB. There is also a recent suggestion that the experimental signature of TRSB is not intrinsic but originates from inhomogeneous strain fields near edge dislocations (108).

The nature of TRSB superconductivity, and the need to understand how it is detected, is a question with a much wider applicability than merely the particular case of Sr_2RuO_4 . μSR experiments, carried out in zero applied magnetic field, are very sensitive to the presence of weak magnetic fields, including those from nuclear moments that cause the muon spin to relax even above T_c . If the development of superconductivity below T_c is accompanied by the appearance of a spontaneous magnetic field, then this can be detected by the appearance of a very small additional relaxation that adds in quadrature with the relaxation due to nuclei and which only appears below T_c (a couple of examples are shown in **Figure 7**). Such an effect is absent for most SCs,

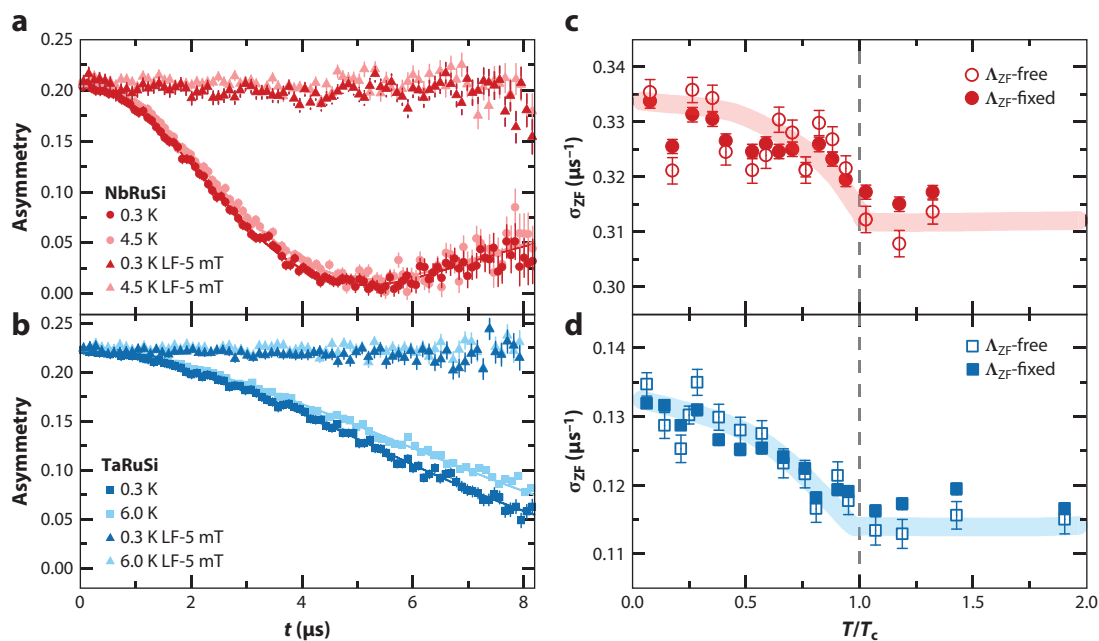


Figure 7

(a,b) Zero-field μSR data for NbRuSi and TaRuSi, with the temperature dependence of the small additional relaxation rate shown in panels c and d. The additional relaxation that sets in below T_c in each case is an extremely tiny additional effect that adds on to the effect of the nuclear relaxation. Nb has a larger nuclear moment than Ta, but the other elements in these compounds (Ru, Si) have zero nuclear moment for the dominantly occurring isotope. Figure adapted from Reference 109 (CC BY 4.0). Abbreviations: μSR , muon spin rotation; LF, longitudinal field; ZF, zero field.

but its presence in experiments has been used to argue for TRSB in various unconventional SCs, including Sr_2RuO_4 (102, 110), LaNiC_2 (111), $\text{PrOs}_4\text{Sb}_{12}$ (112), SrPtAs (113), Zr_3Ir (114, 115), Re (116), Re_6Zr (117), NbRuSi (109) TaRuSi (109, 118), and also KVSb_5 (71) (though in this case, the TRSB is associated with the charge-density wave state, not the superconducting state). There is, however, currently no accepted theory that can predict (a) how large the spontaneous field should be, (b) to what extent these spontaneous fields should be screened by supercurrents, (c) whether these fields are particularly associated with defects, interfaces, and domain boundaries, or indeed (d) whether the presence of the muon itself might play the role of a defect.

In an attempt to examine the effect of muon-induced effects, electronic structure calculations of the muon site in various candidate TRSB materials (119) have demonstrated that the presence of the muon leads to only a limited and relatively localized perturbation to the local crystal structure. In all the materials studied, any small changes to the electronic structure occur several electron volts below the Fermi energy and, therefore, lead only to minimal changes in the charge density on ions close to the muon. These results imply that the muon-induced perturbation alone is unlikely to lead to the observed spontaneous fields in these materials, whose origin is therefore more likely intrinsic to the superconducting state. However, an intriguing possibility is that quantum diffusion of the muon at low temperatures may be important in some materials, as has been shown is probably the case for rhenium (Re), a type I SC for which time-reversal symmetry should be preserved in the superconducting state (120).

The compound that is most central to this discussion is Sr_2RuO_4 , which superconducts below about 1.5 K and has a well-defined q2D electrical structure with three nearly cylindrical Fermi surface sheets (121). There is substantial evidence for the existence of line nodes in the gap function, with these nodes oriented parallel to the c axis (122, 123), and this is consistent with the picture that emerges from a recent transverse-field μSR study (59), the results of which agree extremely well with measurements of the electronic structure (124). The superconducting state in this material can also be tuned using both pressure, in which the TRSB transition in zero-field tracks the superconducting transition measured in transverse field (125), and anisotropic pressure, which shows that the two transitions diverge, which is consistent with the qualitative expectations for a chiral order parameter (75). However, the spontaneous fields that are inferred from zero-field μSR (102) are not detected in surface-scanning measurements (126), and the split into two transitions is not observed in thermodynamic measurements (127, 128). A possible resolution of the situation is that quantum diffusion of the muon (as discussed in the previous section) is taking place in this compound and that the opening up of the superconducting gap is changing, albeit slightly, the diffusion rate and, hence, leading to a small increase in the muon spin relaxation rate. In this scenario, the observed effect would not then be associated with TRSB but would be a consequence only of the opening of the superconducting gap or indeed a change in diffusion rate induced by the application of uniaxial stress. This motivates the search for a more detailed understanding of the stability of muon sites in these TRSB-candidate SCs and how the stability of those sites might be manipulated by both stress and pressure.

7. CONCLUSIONS

This review has demonstrated how implanted muons can be used to probe the magnetic and superconducting properties of a wide range of materials. The technique is expected to continue to play a major role in the future and will be in the front line of measurements of newly discovered advanced materials. To make best use of the technique, it will be necessary to have a greater understanding of implanted muon sites and the effect that they have on local properties, particularly when trying to understand TRSB. This problem can be now tackled using newly developed

density functional methods, so that the coming decade should witness substantial progress being made in this area.

SUMMARY POINTS

1. Muon spin rotation (μ SR), performed using a transverse field, can reveal the field distribution inside vortex lattices, allowing measurement of the penetration depth and vortex lattice dynamics.
2. μ SR can reveal proximal magnetic phases and allow the coexistence of magnetism and superconductivity to be extracted.
3. μ SR, performed in zero magnetic field, can identify spontaneous fields that may be ascribed to the breaking of time-reversal symmetry.
4. Knowledge of the implanted muon site, as well as the stability of that site, may be essential for correct interpretation of some of these phenomena.

FUTURE ISSUES

1. How prevalent is quantum diffusion in oxides, and can some of the observed signatures of the breaking of time-reversal symmetry breaking be ascribed to superconductivity-induced changes in the diffusion rate?
2. Can we better identify and better understand the stability of implanted muon sites in superconductors?

DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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